

Continual Adaptation for Efficient Machine Communication

Robert D. Hawkins¹, Minae Kwon², Dorsa Sadigh^{2,3}, Noah D. Goodman^{1,2}

Departments of ¹Psychology, ²Computer Science, and ³Electrical Engineering
Stanford University, Stanford, CA 94305

{rxdh, mnkwon, ngoodman}@stanford.edu, dorsa@cs.stanford.edu

Abstract

To communicate with new partners in new contexts, humans rapidly form new linguistic conventions. Recent neural language models are able to comprehend and produce the existing conventions present in their training data, but are not able to flexibly and interactively adapt those conventions on the fly as humans do. We introduce an interactive repeated reference task as a benchmark for models of adaptation in communication and propose a regularized continual learning framework that allows an artificial agent initialized with a generic language model to more accurately and efficiently communicate with a partner over time. We evaluate this framework through simulations on COCO and in real-time reference game experiments with human partners.

1 Introduction

Communication depends on shared conventions about the meanings of words (Lewis, 1969), but the real-world demands of language use often require agents to go *beyond* fixed conventional meanings (Grice, 1975; Davidson, 1986). Recent work on *pragmatic* and *context-aware* models has approached this problem by equipping speaker and listener agents with the ability to explicitly reason about one another. Pragmatic reasoning allows listeners to infer richer intended meanings by considering counterfactual alternatives, and allows speakers to be appropriately informative, not merely truthful (Goodman and Frank, 2016; Andreas and Klein, 2016; Fried et al., 2018; Monroe et al., 2017; Vedantam et al., 2017).

These models have largely focused on one-shot settings, where the context is the immediate visual environment. In common interactive settings, however, the relevant context for pragmatic competence also includes the history of previous interactions with the same communication partner. Human in-

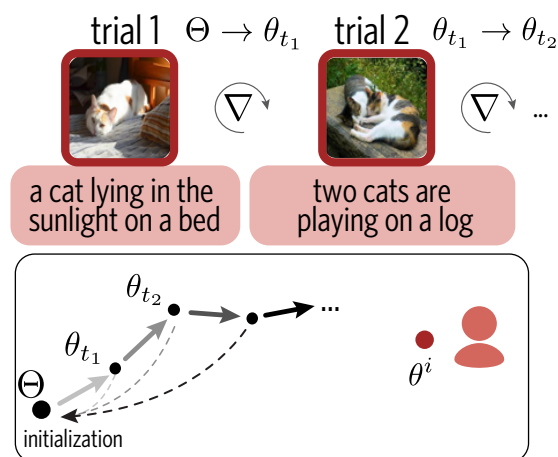


Figure 1: We introduce a regularized continual learning approach allowing agents initialized with a pre-trained language model Θ to iteratively infer the language model θ^i used by a partner, over repeated interactions $\{t_1, t_2 \dots\}$ in an online reference game.

terlocutors are able to establish *ad hoc* conventions based on this history (Clark and Wilkes-Gibbs, 1986; Clark, 1996), allowing for increasingly *accurate* and *efficient* communication. Speakers can remain understandable while expending significantly fewer words (Krauss and Weinheimer, 1964; Orita et al., 2015; Staliūnaitė et al., 2018; Hawkins et al., 2020a; Stewart et al., 2020).

For example, consider a nurse visiting a bed-ridden patient at their home. The first time the patient asks the nurse to retrieve a particular medication, they must painstakingly identify a specific bottle, e.g. “the medicine for my back pain in a small blue medicine bottle labeled Flexeril in my bathroom.” But after a week of care, they may just ask for the “back meds” and expect the nurse to know which bottle they mean. Such flexibility poses a challenge for current pragmatic models. For an artificial agent to establish new conventions, as humans do, it must go beyond pragmatic reasoning at the single-utterance timescale to *learn* about